

The Sum of the Square Roots of 2 and 3 Is Irrational

Formal status: Kernel verified

Faithfulness: User approved; independently reviewed by claude-opus-5 on claude (agreed)

Informal completeness: Not independently assessed

Document status: TeX compiled

Theorem

For every rational number q , the real number q (viewed in $\mathbb{R}$ via the canonical embedding Rat.cast) is not equal to $\sqrt{2} + \sqrt{3}$. Equivalently, $\sqrt{2} + \sqrt{3}$ is irrational.

In Lean: `theorem irrational_sqrt_two_add_sqrt_three : q :  $\mathbb{Q}$ , (q :  $\mathbb{R}$ )  $\neq$  Real.sqrt 2 + Real.sqrt 3`

Approved interpretation

- the real numbers, Mathlib’s ‘Real’
- the rational numbers, Mathlib’s ‘Rat’, mapped into $\mathbb{R}$ by the canonical coercion ‘Rat.cast : $\text{Rat} \rightarrow \mathbb{R}$ ’ (written ‘ $(q : \mathbb{R})$ ’)
- ‘Real.sqrt : $\mathbb{R} \rightarrow \mathbb{R}$ ’ the nonnegative square-root function on $\mathbb{R}$ (value 0 on negative inputs, irrelevant here since $2 > 0$ and $3 > 0$)
- The literals 2 and 3 elaborate as real numbers, forced by the argument type of ‘Real.sqrt’
- One explicit universal quantifier inside the proposition: ‘ $q : \mathbb{Q}$ ’
- The statement is closed (no free variables, no binders in the signature)
- None; the claim is unconditional.
- Standard Mathlib coercion ‘ Rat.cast ’ is available as a ‘RatCast’ instance.
- ‘Real.sqrt 2’ and ‘Real.sqrt 3’ denote the principal nonnegative real square roots, so the sum is the intended real number $\sqrt{2} + \sqrt{3} \approx 3.14626$.
- Repair attempt 2: replaced the Mathlib predicate ‘Irrational’ with its unfolded meaning ‘ $q : \mathbb{Q}, (q : \mathbb{R}) \neq \text{Real.sqrt } 2 + \text{Real.sqrt } 3$ ’. ‘Irrational’ lives in ‘Mathlib.Data.Real.Irrational’ and is not available under a restricted or fixed import set; the unfolded form needs only ‘Real’, ‘Rat’, the ‘ Rat.cast ’ coercion, and ‘Real.sqrt’, so it elaborates under a much weaker import prefix. This is the definitional content of ‘Irrational x’ ($x \in \text{Set.range } ((\cdot) : \mathbb{R} \rightarrow \mathbb{R})$), so nothing is weakened or strengthened.
- Also dropped the dependence on ‘Set.range’ and on set-membership notation, which the previous form pulled in implicitly through ‘Irrational’.

- Used the `-with-` form rather than `¬ q : , ...` to avoid any negation/parenthesization ambiguity when the signature is assembled; the two are logically equivalent.
- `'binders'` is left empty and the quantifier is carried inside `'proposition'`, so the statement remains closed; `'proposition'` contains only a `Prop`-valued term, never a `'theorem'` keyword or colon.
- Removed the explicit `'(2 : )' / '(3 : )'` ascriptions: `'Real.sqrt'`'s argument type already forces the literals to `,` and fewer ascriptions means fewer places for elaboration to fail.
- `'sqrt(2)'` and `'sqrt(3)'` are read as principal (nonnegative) square roots, the standard convention; no other branch is asserted.
- The claim is neither weakened (e.g. to `'not an integer'`) nor strengthened (e.g. to a degree-4 or algebraic-independence claim).

Human-readable proof

Overview. The direct approach `arguing about 2 and 3 separately` is awkward, because a sum of two irrationals can perfectly well be rational (for example $2 + (1 - 2) = 1$). The efficient route is to square the supposed rational value: squaring $2 + 3$ collapses the two surds into the single surd 6, and the resulting identity exhibits 6 as a rational number. Since 6 is irrational, no such rational value can exist.

Setup. Suppose, for contradiction, that there is a rational number q with

$$(q :) = 2 + 3. ()$$

Here $\sqrt{}$ denotes the real square-root function, which on nonnegative arguments satisfies $(x)\sqrt{} = x$.

Step 1: elementary square-root facts. Because $2 \geq 0$ and $3 \geq 0$ we have

$$(2)\sqrt{} = 2, (3)\sqrt{} = 3.$$

Also, because $2 \geq 0$, the multiplicativity of the square root on nonnegative arguments gives

$$2 \sqrt{} 3 = (2 \sqrt{} 3) = 6.$$

(In the formal proof these three facts are supplied by `Real.sq_sqrt` and `Real.sqrt_mul` together with a numeric computation.)

Step 2: square the hypothesis. Squaring both sides of $()$ and expanding the binomial:

$$q\sqrt{} = (2 + 3)\sqrt{} = (2)\sqrt{} + 2\sqrt{}2\sqrt{}3 + (3)\sqrt{}.$$

Substituting the three facts of Step 1 yields

$$q\sqrt{} = 2 + 26 + 3 = 5 + 26. ()$$

Note the point of the manoeuvre: the two individual surds have disappeared, and exactly one surd, 6, remains, appearing linearly.

Step 3: 6 is forced to be rational. Solving $()$ for 6 is pure linear algebra over $:$

$$6 = (q\sqrt{} - 5)/2.$$

Now let $r := (q\check{5})/2$, computed inside $\mathbb{Q}$. This is a legitimate rational number, since $\mathbb{Q}$ is a field and $q \in \mathbb{Q}$. Because the embedding ι is a ring homomorphism, it commutes with squaring, subtraction and division by 2, so the image of r in $\mathbb{R}$ equals $(q\check{5})/2$ computed in $\mathbb{R}$, which is 6. Hence 6 lies in the range of the embedding ι that is, 6 is rational. (Recall that in this formalization "Irrational x " is by definition the statement that x is not in the range of that embedding, so producing the witness r is exactly what is needed to contradict irrationality.)

Step 4: but 6 is irrational. The number 6 is a natural number that is not a perfect square: $2\check{5} = 4 < 6 < 9 = 3\check{5}$, so 6 is strictly between two consecutive squares and therefore is not the square of any natural number. For a natural number n that is not a perfect square, n is irrational. (The classical argument: if $n = a/b$ in lowest terms, then $a\check{5} = n\ b\check{5}$, and comparing the exponent of any prime dividing n to an odd power on the two sides gives a contradiction; equivalently, a rational root of the monic integer polynomial $X\check{5} - n$ must be an integer by the rational root theorem, and no integer squares to 6.) In the formal development this step is discharged by `norm_num`'s irrational-square-root extension, which is exactly the "strictly between consecutive squares" criterion mechanized.

Step 5: contradiction. Steps 3 and 4 assert respectively that 6 is and is not in the range of ι . This is a contradiction, so the assumption $(\)$ is untenable.

Conclusion. No rational number q satisfies $(q : \mathbb{Q}) = 2 + 3$; that is, $2 + 3$ is irrational.

Remarks.

1. Where the hypotheses were used. Nonnegativity of 2 and 3 was needed twice: to get $(x)\check{5} = x$, and to get multiplicativity $2\check{u}3 = 6$. Both would fail without it, so the numeric side conditions are not decorative.
2. No case split on $q = 0$ is required. An alternative proof isolates 2 by writing $3 = q\ 2$ and squaring to obtain $2 = (q\check{5} - 1)/(2q)$, which needs the separate observation that $q \neq 0$ (true since $2 + 3 > 0$). The 6 route avoids this division entirely, which is why it is shorter both on paper and in Lean.
3. Generality. The same argument shows $m + n$ is irrational whenever $m, n \neq 0$ are integers with mn not a perfect square: squaring gives $(mn)\check{5} = (q\check{5} - m - n)/2$, and irrationality of (mn) closes the case. The instance $m = 2, n = 3$ has $mn = 6$, which is not a perfect square.

Status of this exposition. This write-up is a human-readable rendering of the argument; it has not been independently assessed here. The corresponding Lean proof body was reported as accepted by the Lean checker, but a successful tool run is feedback only—acceptance of the solution rests solely with Hardy's FinalVerifier.

Known gaps

- This informal exposition has not been independently assessed. It is a human-readable rendering of the formal argument, not a verified object in its own right; any discrepancy between this prose and the Lean proof body is possible in principle.
- A successful Lean tool run is feedback only. Only Hardy's FinalVerifier can accept the solution, and no such acceptance is claimed here.
- Step 4 (irrationality of 6) is not proved from first principles in the formal development: it

is discharged by Mathlib’s `norm_num` irrational-square-root extension, which mechanizes the criterion ‘ $k\sqrt{n} < n < (k+1)\sqrt{n}$ implies n is irrational’. The classical justification sketched in the prose (rational root theorem / prime-exponent parity) is offered as explanation, not as the formally checked route, and the correctness of that extension is taken on trust as part of the pinned Mathlib.

- Standard facts used without separate proof, as library lemmas of the pinned Mathlib: $(x)\sqrt{} = x$ for $x \geq 0$ (`Real.sqrt_sqrt`), $(xy)\sqrt{} = x\sqrt{}y$ for $x \geq 0$ (`Real.sqrt_mul`), the ring-homomorphism property of the coercion `↑`, and the definitional unfolding of `Irrational` as non-membership in the range of that coercion.
- Steps 2 and 3 are carried out in the formal proof by the automation tactics `nlinarith`, `push_cast` and `linarith` rather than by explicitly written algebraic manipulations; the human-readable expansion of the binomial and the linear solve given above reconstruct what those tactics are expected to establish, but the exact certificates they produced were not inspected.
- The generalization stated in Remark 3 ($m + n$ irrational when mn is not a perfect square) is asserted as context only. It was not formalized or checked, and lies outside the Frozen Claim.

Exact Lean statement

```
theorem irrational_sqrt_two_add_sqrt_three : q : ℚ, (q : ℝ) < Real.sqrt 2 + Real.sqrt 3
```

Verification appendix

Axioms used: `propext`, `Classical.choice`, `Quot.sound`

```
Frozen Claim SHA-256: 67bbf6be3f6cce3fbcf0c18652aeedf6f61ed5d0c09a91ebedbfd36aa4077204
Lean source SHA-256: 7280e5b6de3e27f93d0d41605a25aafda4c36fb3d8955945633cd93c5cbebec3
Verification SHA-256: 35f53ddb64ff223571daedcd425260158466ce1e30ff2a44144b13088e4b2f9
Prompt set SHA-256: 50c207a708e3696c9846957a60089ccd0fb4a6ddbc6a6920449887455cf3f5e2
Tectonic bundle SHA-256: 6ffe055852f8faf66c0acbe1a7fb27f87b869a90bad1204f3bf4d9683f597c7c
```

Run and tool identities

- Run ID: 3c689084-8c38-41f6-ad81-c3bf4130b0ce
- Model: claude-opus-5
- Backend: claude
- Runtime SDK: 0.2.127
- Lean: 4.33.1
- Mathlib: 0df444a360eaa60ab8c11dca51a86af692955474
- Tectonic: 0.16.9
- Tectonic bundle: <https://data1.fullyjustified.net/tlextras-2022.0r0.tar>

Lean kernel verification, faithfulness (approved by the user and read back by an independent model before proof search), heuristic informal review, and document compilation are independent statuses. This trusted-development MVP does not sandbox generated Lean or TeX.